

Tharunikha P A

📍 Karur, India | ✉ tharunikhaannadurai@gmail.com | ☎ +91 6380744406 | in [Tharunikha](#) | 🌐 [tharunikhaa](#)

Professional Summary

Results-driven ECE student with strong interest in telecommunications, software development, and intelligent systems. Skilled in Java, C, Python, JavaScript, Node.js, MySQL, C, and Full Stack Java development, with hands-on experience in building IoT-based smart solutions and real-world engineering applications. Passionate about leveraging technology, networking concepts, and innovative problem-solving to develop scalable and impactful solutions in the telecom and digital communication industry.

Education

B.E in Electronics and Communication Engineering

M. Kumarasamy College of Engineering – CGPA: 8.14/10

2023 – 2027 (Current)

Higher Secondary Education (HSC)

Little Flower matric hr.sec.school – Percentage: 84.166

2022 – 2023

Secondary School Leaving Certificate (SSLC)

Little Flower matric hr.sec.school – Percentage: 80

2019 – 2020

Area of Interest

JAVA programming , DBMS

Skill Set

Technical Skills:

- **Programming Languages:** C, JAVA, Python, Data Structure, PHP, HTML
- **Frontend Technologies:** HTML, Bootstrap 5, DataTables
- **Databases:** MySQL, Node.js
- **Tools and Platforms:** Git, GitHub, VScode, EDA playground, MATLAB, LT spice
- **Visualization and Reporting:** Chart.js, Excel/PDF export
- **Hardware tools:** network analysis, Embedded system, IoT design, MATLAB simulation

Soft Skills:

- Communication, Interpersonal Skills, Stakeholder Coordination
- Interview Coordination, Follow-up Management, Team Collaboration
- Problem-Solving, Time Management

Experience

Full Stack Developer

Technology Innovation Hub (TIH), M. Kumarasamy College of Engineering | Karur, India

Aug 2025 – Apr 2026

- Architecting an RBAC layer across 5+ departments using granular middleware to enforce secure data iso-lation between Students, Mentors, Advisors, and HODs.

Solution Developer (VLSI)

Taras Systems and Solutions, VLSI (Verilog and Sys-verilog) | Online

Dec 2025 – May 2026

- In the internship , I have learned about the VLSI design code implementation and also done a project under VLSI major “A Reconfigurable system for smart sensor unit” also attended conference for the project. It is a online internship , but learned with full interest.

Projects

Discipline Management System | [GitHub](#)

Tech Stack: PHP, MySQL, HTML, CSS, JavaScript, JAVA

- Developed a web-based Discipline Management System to efficiently monitor and manage student behavior, violations, and disciplinary records.
- Designed modules for recording incidents, assigning penalties, and maintaining a structured history of student discipline data.
- Implemented role-based access for administrators and staff to ensure secure data handling and controlled system usage.

Hybrid Industrial Wastewater Management System using IoT | [GitHub](#)

Tech Stack: ESP32, IoT Sensors, Embedded C, Python, OpenCV, ThingSpeak, JAVA, HTML, CSS, JavaScript

- Developed an IoT-based hybrid industrial wastewater management system for real-time monitoring and treatment of contaminated water.
- Integrated TiO₂ filtration, algae microbial fuel cell, biosensors, and AI-based monitoring to improve water purification efficiency and support sustainable water reuse.
- Implemented sensor-based water quality analysis with cloud monitoring and automated data visualization for industrial applications.
- Filed an Intellectual Property (IP) Patent for the innovative hybrid wastewater treatment and smart monitoring system.

An Efficient low-power reconfigurable system design for smart signal processing

Tech stack: VLSI Verilog code, system Verilog code, EDA playground

- Developed a low-power reconfigurable system architecture for smart signal processing applications aimed at improving processing efficiency and reducing power consumption in embedded systems.
- The project focused on implementing an adaptable hardware design capable of supporting real-time signal processing with enhanced performance and energy optimization.
- I have completed the conference paper for our project and also published in the IEEE xplore.

Certifications

- Certified system administrator - ServiceNow
- Certified Application Developer - ServiceNow
- Conference attended certificate for the project Reconfigurable signal processing system
- Machine Learning – Understanding data-driven models and intelligent systems
- Generative AI – Exploring the foundations of AI-powered content creation
- Cloud 101 – Building strong fundamentals in cloud computing with AWS
- Certified in Network Security - NPTEL

Personal Information

Father's Name : S.Annadurai

Date of Birth : 19.06.2006

Blood group : O+

Gender : Female

Hobbies : Gardening, listening music

Nationality Language's proficiency: Tamil (native proficiency), English (professional proficiency)